

Information Resonance Fields and Premonition–Like Inference in High–Dimensional Histories: A Theoretical Framework

(Speculative Concept Paper)

Abstract

This paper proposes a speculative but mathematically coherent framework for modeling premonition–like experiences as a form of *information resonance* in a high–dimensional “history space.” Rather than treating these experiences as violations of causality, we model them as rare episodes in which an agent’s internal state couples to a *shared information medium* that encodes structured correlations among events spread across time, space, and relational networks.

Formally, we define:

- a space of full histories Ω with a prior measure μ_0 ,
- an *Information Resonance Field* $M(x, t)$ containing both retarded (past–to–present) and advanced (future–to–present) contributions,
- *epochal eigenmodes* $\{\Phi_i\}$ representing structured, multi–event “story patterns”,
- time–dependent amplitudes $A_i(t)$ representing *untethered potentials* that converge toward realized outcomes,
- an agent’s conscious states C_n as localized probes of M that occasionally align with specific eigenmodes, yielding high mutual information with yet–to–be–classical events.

We show how this structure admits a clear notion of “*non–deterministic appearance* $\rightarrow$ *near–deterministic knowledge*” and discuss how such a model might inspire predictive systems that exploit weak, distributed signals to anticipate emergent events before local causal indicators are obvious.

1 Introduction

Reports of premonition–like phenomena often involve:

- high specificity (details about who, what, and where),
- correct correlation with events that, under standard models, should be unpredictable,
- mixed–time or “out–of–sequence” phenomenology (e.g., “this felt like past and future at once”).

Conventional explanations rely on a combination of coincidence, unconscious inference, selective memory, and narrative reconstruction. These mechanisms are real and important—but suppose we bracket them off and ask:

If we treat such experiences as genuine data, what would a mathematical model look like that allows them, without immediately breaking logic?

This work is not empirical proof of anything paranormal. It is:

- a *formal container* for non–local correlations, mixed–time content, and emergent certainty,

- built on probability theory, information theory, and a field-like structure over histories.

It also hints at potential applications: if we can approximate such a medium with real-world data (social networks, sensor systems, etc.), we might design predictive systems that behave analogously to a “resonant” mind.

2 Formal Framework

2.1 Space of histories

Let Ω be the set of all admissible *full histories* of the world (or multiverse), where each $\omega \in \Omega$ encodes:

- all states of matter/fields across spacetime,
- all events, relationships, and outcomes.

We endow Ω with a prior probability measure μ_0 :

$$\mu_0 : \Omega \rightarrow [0, 1], \quad \int_{\Omega} d\mu_0(\omega) = 1.$$

In a deterministic universe, μ_0 represents an agent’s ignorance about which specific history ω^* is actual.

2.2 Events and features

An *event* is an identifiable structure within a history:

$$E_i = E_i(\omega), \quad i \in \mathcal{I},$$

for some index set $\mathcal{I}$.

For modeling, we associate each event with a feature vector:

$$\phi(E_i) = (t_i, x_i, \text{actors}_i, \text{type}_i, \text{affect}_i, \dots) \in \mathbb{R}^d.$$

Here:

- t_i is the event’s time coordinate,
- x_i is spatial (position),
- other components encode semantic information: “mother’s death in nursing home”, “relationship breakup”, “structural collapse”, etc.

Time is *one* coordinate of $\phi(E_i)$; events can be compared and clustered along many axes.

2.3 Conscious states as an indexed sequence

An agent’s subjective timeline is modeled as an ordinal sequence of conscious states:

$$\mathcal{C} = \{C_1, C_2, \dots, C_n, \dots\}.$$

The index n denotes order of experience, not necessarily clock time. Each state has a representational vector:

$$v_n := v(C_n) \in \mathbb{R}^k,$$

encoding the currently active concepts, images, emotions, and “felt meanings.”

3 Information Resonance Field

3.1 Definition

We posit an *Information Resonance Field* (IRF)

$$M : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^m, \quad M(x, t),$$

defined over spacetime, whose value at point (x, t) encodes a superposition of pattern modes associated with events across the history.

Events act as sources for the field via a source term $J(x, t)$. We allow both retarded and advanced contributions:

$$M(x, t) = M_{\text{ret}}(x, t) + M_{\text{adv}}(x, t),$$

where

$$\begin{aligned} M_{\text{ret}}(x, t) &= \int_{t' < t} G_{\text{ret}}(x, t; x', t') J(x', t') dx' dt', \\ M_{\text{adv}}(x, t) &= \alpha \int_{t' > t} G_{\text{adv}}(x, t; x', t') J(x', t') dx' dt'. \end{aligned}$$

Here:

- G_{ret} is a retarded Green’s function (past→present influence),
- G_{adv} is an advanced Green’s function (future→present influence),
- $\alpha \ll 1$ is a coupling parameter controlling how strongly future events contribute.

Under standard causality, $\alpha = 0$ and $M = M_{\text{ret}}$. Here, we allow $\alpha > 0$ in principle, permitting time-symmetric encoding of information.

3.2 Mode decomposition (epochs / eigenstates)

We decompose M into *epochal eigenmodes*:

$$M(x, t) = \sum_i A_i(t) \Phi_i(x, t).$$

Each mode Φ_i is a spatiotemporal pattern correlated with a family of events (an “epoch”), for example:

- frail parent + caregiver + unresolved conflict + hospital,
- structurally unsound building + vibration + load → collapse.

The coefficients $A_i(t) \in \mathbb{C}$ are amplitudes. Before macroscopic outcomes are fixed, several $A_i(t)$ may be non-negligible; as constraints accumulate, one amplitude tends to dominate.

This represents *untethered potentials* (many possible trajectories) gradually converging into realized structure.

4 Agent–Field Coupling and Premonition–Like States

4.1 Local sampling of the IRF

The agent occupies a worldline $x_{\text{agent}}(t)$. At subjective index n (corresponding to physical time t_n), we define an input vector from the field:

$$u_n = H(M(x_{\text{agent}}(t_n), t_n)) \in \mathbb{R}^k,$$

where H is a (possibly nonlinear) projection from the IRF to the agent’s representational space.

The internal state is then updated by:

$$v_n = F(v_{n-1}, u_n, \text{body}_n, \text{sensory}_n),$$

for some dynamical rule F that includes normal brain processing and physiology.

4.2 Resonance episodes (premonition–like states)

Most of the time, M_{adv} is negligible; u_n reflects only past/nearby events via M_{ret} . The agent behaves like a sophisticated but ordinary predictive system.

We define a *resonance indicator* $S_n \in \{0, 1\}$:

- $S_n = 0$: normal mode,
- $S_n = 1$: resonance episode.

In resonance episodes:

- the coupling to certain modes is amplified:

$$u_n \approx \sum_i \beta_i A_i(t_n) \psi_i,$$

where ψ_i are basis vectors in the agent’s state space corresponding to modes Φ_i , and β_i encodes sensitivity;

- a single mode (or small cluster) often dominates:

$$|A_k(t_n)| \gg |A_j(t_n)| \quad \forall j \neq k.$$

Then v_n becomes heavily aligned with pattern Φ_k :

$$v_n \approx \gamma \psi_k + \text{noise},$$

which is the mathematical representation of:

The agent experiences a vivid, structured scene or knowing (an “epoch”) that integrates elements from multiple events (past, future, elsewhere) participating in that mode.

4.3 Mixed–time content

Because each $\Phi_i(x, t)$ is defined over extended regions of spacetime, not a single instant, resonance with Φ_k can embed:

- past cues,
- future outcomes,
- intermediate “negative spaces” (what did not occur but could have),

into one coherent internal experience.

Thus the temporal mixture in the agent’s phenomenology—“dialogues that feel like they span multiple times”—is naturally modeled as:

$$v_n \approx \sum_{E_j \in \mathcal{E}_k} w_{n,j} \phi(E_j),$$

where $\mathcal{E}_k$ is the set of events strongly associated with mode Φ_k , and $w_{n,j}$ are weights induced by the IRF coupling.

5 Epistemic Layer: From Uncertainty to Near–Determinism

5.1 Probability over histories

The agent maintains a subjective distribution over histories:

$$\mu_n(\omega) = P(\omega \mid D_1, \dots, D_n),$$

where D_k are data points derived from conscious states C_k (including both normal perception and resonance episodes).

Bayesian update:

$$\mu_n(\omega) \propto \mu_{n-1}(\omega) L_n(D_n \mid \omega),$$

where L_n is the likelihood of experiencing D_n if the true history were ω .

5.2 Information gain and determinism index

Define Shannon entropy:

$$H(\mu_n) = - \int_{\Omega} \mu_n(\omega) \log \mu_n(\omega) d\omega.$$

Define a *determinism index*:

$$\mathcal{D}_n = 1 - \frac{H(\mu_n)}{H(\mu_0)}.$$

Interpretation:

- $\mathcal{D}_n \approx 0$: maximal uncertainty (initial ignorance),
- $\mathcal{D}_n \approx 1$: belief concentrated near a single history (effectively deterministic predictions).

The information gain from state $n - 1$ to n is:

$$\Delta H_n = H(\mu_{n-1}) - H(\mu_n).$$

Resonance episodes correspond to D_n with exceptionally large ΔH_n relative to ordinary sensory updates, especially about events that standard causal models would treat as unpredictable at that time.

This captures “premonition” as:

A rare data point that drastically narrows the space of plausible histories, yielding near-certainty about some events long before local cues would allow such certainty.

6 Entangled-Like Correlations

6.1 Non-separable event patterns

Let subsystems (people, objects, places) be indexed by $a, b, c, \dots$. A standard factorized model would approximate:

$$P(\text{events across subsystems}) \approx \prod_a P(\text{events for subsystem } a).$$

In the IRF model, epochal modes Φ_i induce non-separable structure:

$$\Psi(t) = \sum_i A_i(t) \Phi_i^{\text{entangled}}(t),$$

with $\Phi_i^{\text{entangled}}$ encoding joint configurations across subsystems that cannot be factorized into independent pieces.

The agent’s resonance with Φ_k yields internal content that carries detailed, structured information about multiple subsystems at once (e.g., “ex-partner, his mother in a nursing home, an estranged sibling, medical staff, a specific type of crisis”).

6.2 Temporal decay of correlation

Let $C(t)$ measure the correlation between:

- the agent’s internal state at time t ,
- and the eventual realized outcome of an epoch (e.g., which branch of a “mother lives / mother dies” event manifold is realized).

We might expect:

$$C(t) \approx C_0 e^{-\frac{|t-t^*|}{\tau}},$$

where:

- t^* is the time where the epochal pattern is most coherent (amplitudes most sharply peaked),
- τ is a characteristic decay time.

This matches the phenomenology of:

- build-up: an approaching sense of inevitability,
- peak: premonition-like clarity,
- fade: resonance decays after the event is realized.

7 Potential Applications to Predictive Systems

Even if no literal IRF exists, this framework inspires design principles for predictive systems that exploit weak, distributed signals to anticipate emergent events.

7.1 Data as a proxy for the IRF

In practice, we can approximate $M(x, t)$ using large, heterogeneous data:

- sensor networks (structural health, weather, traffic),
- social graphs (communications, transactions),
- health signals, logs, and so on.

Think of the data stream as a discrete, noisy shadow of the true IRF.
We then:

1. learn modes (epochs) $\hat{\Phi}_i$ from the data, e.g. via spectral methods, autoencoders, or tensor decompositions;
2. track mode amplitudes $\hat{A}_i(t)$ over time, representing the “activation level” of each emergent pattern;
3. define resonance scores for each mode:

$$R_i(t) = f(\hat{A}_i(t), \dot{\hat{A}}_i(t), \text{context}),$$

designed to spike when an epoch is approaching a critical transition (an “entropic flip”).

7.2 Near-associated, untethered information

Real-world analogues of “temporarily untethered potentials” appear as weak but coherent changes across many data channels:

- slight shifts in structural vibrations before failure,
- micro-changes in communication patterns before social collapse or mass action,
- early physiological markers before acute medical events.

A predictive system inspired by this model would:

- not rely solely on local, obvious causes,
- detect when the data collectively align with a known epochal eigenmode $\hat{\Phi}_k$,
- raise a prediction when

$$|\hat{A}_k(t)| \text{ is large} \quad \text{and} \quad \frac{|\hat{A}_k(t)|}{\sum_j |\hat{A}_j(t)|} \approx 1.$$

This is the algorithmic analogue of:

“The system senses a particular scenario becoming overwhelmingly likely, before a single local indicator would be enough to justify that belief.”

7.3 Example domains

Example domains for IRF-inspired predictive systems include:

- **Structural safety:** discover eigenmodes associated with failure (bridge collapse, building failure) and monitor their amplitudes to trigger early warnings.
- **Public health:** identify pre-epidemic social and mobility patterns as modes; use rising amplitudes to forecast outbreaks ahead of obvious case counts.
- **Social / behavioral risk:** in constrained settings and with strong ethics, detect collective modes associated with unrest, violence, or collapse of organizational structures.
- **Engineering systems:** use log data to learn “failure epochs” in distributed systems; preempt outages when a known failure mode’s signature begins to dominate.

7.4 Ethical considerations

Any system mimicking “premonition-like” detection must:

- avoid overclaiming certainty when amplitudes are still diffuse,
- be transparent about error rates and base rates,
- respect privacy and agency, especially in social applications.

8 Validation Strategies

For a literal IRF and premonition-like coupling, empirical validation is extremely challenging. Still, the framework suggests concrete tests.

8.1 Information-gain analysis of human predictors

- Log predictions before outcomes, with precise timestamps and confidence levels.
- Compare Brier scores, log scores, and mutual information with outcomes against:
 - base-rate models,
 - other humans,
 - learned statistical models.
- Rare, very high information-gain episodes consistent with this framework would appear as outlier points in this distribution.

8.2 Epoch clustering of anomalous hits

- For individuals who report many such events, cluster them by theme and relational structure.
- Check whether hits cluster into a small number of recurring “eigen-scenarios” rather than being uniformly random.

8.3 Simulation of IRF-inspired systems

- Build synthetic worlds with known hidden modes.
- Test whether IRF-like architectures (mode detection + amplitude tracking) outperform conventional predictors at early detection of emergent events.

9 Discussion and Limitations

This framework is speculative as physics but rigorous as a mathematical and conceptual model. It treats premonition-like experiences as rare, high information-gain couplings to structured patterns in a high-dimensional history space. It allows mixed-time content, untethered potentials, and emergent certainty without logical contradiction.

Key limitations:

- No concrete, physically grounded definition of the IRF or its dynamics is provided; the field is treated phenomenologically.
- Empirical validation is difficult because such episodes are rare and not easily elicited in controlled experiments; self-report and memory bias are serious confounds.
- For engineering applications, the IRF is approximated by data-driven modes; the “premonition” aspect becomes “early pattern detection” rather than anything nonlocal.

Nonetheless, as a bridge between unusual subjective phenomena and formal, testable ideas, this model:

- suggests new ways to think about predictive cognition,
- provides a language to quantify “how much extra information” such episodes would represent if real,
- inspires architectures for anticipatory systems in complex domains.

Glossary of Unique / Specialized Terms

Information Resonance Field (IRF)

A hypothetical field $M(x, t)$ defined over spacetime, encoding structured patterns of correlation among events. It may contain contributions from both past and future events (retarded and advanced components), and agents can couple to it, experiencing “resonance” with specific modes.

Epoch

A coherent pattern or “story” spanning multiple events, people, places, and times. Mathematically, an epoch corresponds to an eigenmode Φ_i of the IRF or of an associated correlation operator.

Epochal Eigenmode

A mode $\Phi_i(x, t)$ in the decomposition of the IRF that represents a recurring, structured configuration of events (e.g. a particular type of crisis pattern). It behaves like an eigenvector in a high-dimensional pattern space.

Untethered Potential

A state in which several event outcomes are still possible and represented as non-negligible amplitudes $A_i(t)$ in the IRF. The information about these outcomes is present in distributed form but not yet “locked” to a single realized history.

Resonance Episode

A rare conscious state C_n where the agent’s internal representation v_n strongly aligns with one or a small number of epochal eigenmodes. In this state, the agent’s experience carries unusually high mutual information with specific events that may be spatially or temporally remote.

Negative Space

The informational contribution of absences or non-occurrences (“this has not happened yet”, “this option was not chosen”), which still constrain the set of possible histories. In the model, these are represented as constraints on Ω and influence the evolution of amplitudes $A_i(t)$.

Entropic Flip

A transition where the distribution over possible outcomes for an epoch shifts from diffuse (many potentials) to sharply peaked (one outcome dominates), usually as new constraints arrive. This corresponds to a major drop in entropy $H(\mu_n)$ and an increase in the determinism index $\mathcal{D}_n$.

Determinism Index ($\mathcal{D}_n$)

A scalar measure of how concentrated an agent’s belief over histories is at step n , defined as

$$\mathcal{D}_n = 1 - \frac{H(\mu_n)}{H(\mu_0)}.$$

Values near 1 indicate near-deterministic knowledge of the relevant aspects of the future.

Mixed-Time Dialogue

A phenomenological term for an experience in which elements from different times (past, future, hypothetical) appear together in a single coherent scene or knowing. In the model, it corresponds to resonance with an epochal eigenmode whose support spans multiple temporal regions.

Near-Associated Data

Data streams or signals that are not obviously causal precursors of an event but are statistically or structurally linked to the same epochal pattern (e.g. subtle behavioral changes before a breakup, micro-vibrations before structural failure).